National University of Computer and Emerging Sciences

Information Retrieval (CS4051)

Date: April 07, 2025 (9AM-10AM)

Course Instructors

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Sessional-II Exam

Total Time: 1 Hour

Total Marks: 40

Total Questions: **03**

Semester: Spring-2025

Campus: Karachi

Department: AI & DS

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Student Name	Roll No	Section	Student Signature

CLO # 1: <Understand the basic concepts and techniques in Information Retrieval.>
CLO # 4 <Appreciate the importance of data structures, indexes and retrieval efficiency>

Q1. Answer the following questions to the point. **[Time: 30 mins] [Marks: 2x10]**

a. Why is the evaluation of ranked retrieval different from unranked retrieval? Explain.

Ranked retrieval orders documents by relevance, so evaluation considers document position using metrics like Precision@k, MAP, or NDCG. Unranked retrieval treats documents as simply relevant or not, so binary measures like precision and recall are sufficient.

b. Can the tf-idf score of a term in a document exceeds 1?

Yes, tf-idf can exceed 1 depending on how term frequency (tf) and inverse document frequency (idf) are calculated, especially when using raw tf or log-scaled tf with rare terms.

c. What do we mean by Static Quality Scores for a document- (g(d))? Explain.

g(d) is a pre-assigned score reflecting a document's intrinsic quality, such as PageRank or authority, independent of the query.

d. Define cluster pruning. How does it facilitate efficient relevance score computation?

Cluster pruning groups similar documents into clusters. At query time, only cluster leaders are compared with the query. Then, a small subset of matching clusters is explored, reducing computation.

e. Why does relevance feedback not work effectively for web searches? Provide two reasons.

1. Users rarely provide explicit feedback (e.g., marking documents relevant/non-relevant).
2. Queries are often vague or short, making feedback less reliable or effective.
3. Feedback may unintentionally bias results toward already popular or over-optimized content.

f. How can we configure the "More pages like this" option using the Rocchio algorithm?

To configure the "More pages like this" option using the Rocchio algorithm, we treat the current document as a relevant example and build a new query vector from it. The formula becomes:

$$Q = \beta.d$$

This new vector acts like a pseudo-query to retrieve documents similar to the current one. Optionally, non-relevant documents can be subtracted if known. This helps in finding related pages without user input.

g. What are the main assumptions of probabilistic Information Retrieval (IR) models? Provide any two.

1. Binary relevance: Each document is either relevant or not.
2. Term independence: Occurrence of terms is statistically independent given relevance.

h. Describe a scenario where a low fallout is crucial for IR effectiveness.

In medical diagnosis or legal document retrieval, retrieving irrelevant documents (false positives) is risky, so low fallout (few non-relevant documents retrieved) is essential.

i. Suppose a system gives too much weight to a negative feedback. What potential problems could arise?

1. Relevant documents similar to negatively marked ones may be unfairly penalized, reducing retrieval quality.
2. **Query drift:** The query vector may move too far from the user's original intent.

j. Given:
P(Relevant) = 0.4)
P(Query Term | Relevant) = 0.7
P(Query Term | Non-Relevant) = 0.2
Use Bayes' Theorem to compute P(Relevant | Query Term).

We know that from Bayes' Theorem,

P(Relevant/ Query Term) = [P(Query Term / Relevant) x P(Relevant)] / P(Query Term)

The probability of Query Term can be found as:

P(Query Term) = [P(Query Term / Relevant) x P(Relevant)] +
[P(Query Term / Non-Relevant) x P(Non-Relevant)] ……Eq B

As per given,
P(Relevant) = 0.4
So,
P(Non-relevant) = 1- 0.4 = 0.6

Putting the values in Eq B, we get

P (Query Term) = [07) x 0.4] + [P(0.2) x P(0.6)] = 0. 4

Now using Bayes Theorem
P (Relevant/ Query Term) = [0.7 x 0.4] / (0.4) = 0.7

CLO # 2: <Understanding how IR Model works>

Q2. Consider the following documents in a collection. **[Time: 15 mins] [Marks: 10]**

Vocabulary = {w1 w2 w3 w4 w5}, where the number indicates the order of dimensions:
d1= {w1 w2 w2 w1 w5}
d2= {w3 w2}
d3= {w1 w2}
Relevant Documents = d1, d3
Non-Relevant Document= d2
q = {w1 w2 w3}
Using the Probability Ranking Principle (PRP) rank these documents. Assume equal Prior. You can use the Laplace smoothing version as below:

$N_1 = P(w_i \mid R) = (count(w_i \text{ in } R) + 1) / (|R| + |V|)$
$N_0 = P(w_i \mid \neg R) = (count(w_i \text{ in } \neg R) + 1) / (|\neg R| + |V|)$

Let's construct a table, where we compute probability of each words for relevant and non-relevant classes
|V| = 5
Equal Priors implies P(R) = P(~R) =0.5 but we will use |R|=2 and |~R|=1

Word	W1	W2	W3	W4	W5
N_1	0.33	0.33	0.083		
N_0	0.143	0.286	0.286		

As the target classification documents contains only first three words, no need to fill the complete table. Now for RSV(di) we will have

RSV(d1) = (0.33/0.143) * (0.33/0.286) = 2.71

RSV(d2) = (0.33/0.286) * (0.33/0.286) = 0.34

RSV(d3) = (0.33/0.143) * (0.33/0.286) = 2.71

Hence ranking order will be : d1=d3, d2

CLO # 3: <Appreciate the importance of data structures, indexes and retrieval efficiency>

Q3. Consider the table of a retrieved results given below: **[Time: 15 mins] [Marks: 2x5]**

ith	**Rel(ith)**	**Log2(i+1)**	**Rel(ith) / Log2(i+1)**
1	3	1	3
2	3	1.585	1.89
3	2	2	1
4	0	2.32	0
5	1	2.58	0.387

Where Rel(ith) give relevance score for the document; Log2(i+1) give the log position and Rel(ith) / Log2(i+1) represents logarithmic scale of reduction.

Compute the following:

a. Calculate the Cumulative Gain (CG@5)?
CG@5=3+3+2+0+1=9

b. Calculate the Discount Cumulative Gain (DCG@5)?
DCG@5=3.000+1.893+1.000+0+0.387=6.28

c. Calculate the Idealized Discount Cumulative Gain (IDCG@5).
To compute IDCG, sort the relevance scores in descending order:
Rel scores = [3, 3, 2, 1, 0]

i	Rel	$\log_2(i+1)$	Rel / $\log_2(i+1)$
1	3	1	3.000
2	3	1.585	1.893
3	2	2	1.000
4	1	2.32	0.431
5	0	2.58	0.000

IDCG@5=3+1.893+1+0.431+0= 6.324

d. Calculate the Normalized Discount Cumulative Gain (nDCG@5).
nDCG@5= IDCG@5 / DCG@5 = 6.324/6.28 = 0.993

e. Outline two limitations of Normalized Discount Cumulative Gain.
1. Assumes Relevance Scores Are Accurate
2. Sensitive to Log Base & Ranking Depth

Formulas for reference:

$CG_p = \sum_{i=1}^{p} rel_i$	$DCG_p = \sum_{i=1}^{p} \frac{rel_i}{\log_2(i+1)}$	$nDCG_p = \frac{DCG_p}{IDCG_p}$

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